

Gaussian Perturbed Maxima Without Sampling

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Abstract

Softmax is the gradient of a perturbed maximum: independent Gumbel noise turns the multiclass maximum into log-sum-exp. General perturbations give other smooth choice maps and losses, but their expectations are evaluated by Monte Carlo, and the leading heteroscedastic image classifiers, whose output layer is exactly a low-rank Gaussian perturbed argmax, approximate their probabilities by sampling a softmax relaxation. For Gaussian perturbations with low-rank-plus-diagonal covariance we make the whole object explicit and sampling-free. One shared conditional field yields every winner probability, the potential, exact Jacobian–vector products, and, by a new reverse-mode formula, the gradients with respect to means, factor loadings, and idiosyncratic variances, all at cost linear in the number of alternatives. The dual regularizer’s curvature is the effective resistance of a photo-finish graph, with softmax the special case whose conductances are $p_i p_j$. The likelihood and Fenchel–Young objectives separate outside that special case, and learning covariance by the latter is degenerate.

Keywords: perturbed optimizers; multinomial probit; Fenchel–Young losses; graph Laplacian; heteroscedastic classification; effective resistance

1 Introduction

A large class of neural operations is organized around the maximum of N competing scores. The maximizer is discrete, so it is smoothed by perturbation: for noise ξ , define

$$G(\mu) = \mathbb{E} \left[\max_i (\mu_i + \xi_i) \right], \quad p(\mu) = \nabla G(\mu) = \mathbb{E} [e_{\arg \max_i (\mu_i + \xi_i)}].$$

For independent Gumbel noise G is log-sum-exp and p is softmax, up to an additive constant. Berthet et al. (2020) develop this perturbed-optimizer view in generality, including the convex dual and the induced Fenchel–Young loss (Blondel et al., 2020), and evaluate general perturbations by Monte Carlo; they demonstrate Gaussian-perturbed argmax training on CIFAR-10.

The Gaussian member is the natural carrier of correlation. We study

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \epsilon_i, \quad f \sim N(0, I_k), \quad \epsilon_i \sim N(0, 1) \text{ iid},$$

so the perturbation covariance is $\Sigma = VV^\top + \text{diag}(D)$, low-rank plus diagonal. This structure arises automatically from uncertain representations: if a feature vector is $h = \bar{h} + Af$ and scores are $U_i = w_i^\top h + b_i + \sqrt{D_i} \epsilon_i$, then $v_i = A^\top w_i$.

*Working draft. Every identity below is verified numerically (experiment 27 of the companion repository, <https://github.com/microprediction/kinetics>); the neural experiments of Section 7 are pending.

The same model already ships in computer vision. Collier et al. (2021) place exactly this latent-utility argmax, with input-dependent low-rank-plus-diagonal covariance, on the final layer of ImageNet-scale classifiers; lacking a closed form, they approximate the winner probabilities by Monte Carlo over a temperature-softmax relaxation, and the scaled follow-up (Collier et al., 2023) identifies the latency and memory cost of that sampling as a standing issue.

A numerical companion (Cotton, 2026) supplies the missing primitive. Conditional on the factors the alternatives are independent, so all N winner probabilities come from a single product field in $O(QNL)$ operations for Q factor nodes and L lattice points, with exact Jacobian–vector products from the same pass, a weighted graph-Laplacian Jacobian, and a globally invertible map from mean utilities modulo translation to the simplex interior.

This paper asks what that primitive becomes as a neural layer. The novelty boundary is deliberately narrow. Perturbed-max smoothing, its duality, and Fenchel–Young losses are prior art (Berthet et al., 2020; Blondel et al., 2020); generalized-entropy conjugates of random-utility surpluses are prior art (Fosgerau et al., 2020). What we add is what becomes computationally and geometrically explicit in the structured Gaussian case:

1. **The potential is free.** The field that computes p computes G with $O(QL)$ additional work (Section 3).
2. **Dual graph geometry.** The dual regularizer’s Hessian is the pseudoinverse of the photo-finish Laplacian, so dual curvature is effective electrical resistance; softmax is the special graph with conductances $p_i p_j$ (Section 4).
3. **Full reverse mode.** A new shared-field vector–Jacobian product gives gradients in (μ, V, D) at the cost of one forward pass (Section 5).
4. **Two losses, one special case.** The Gaussian Fenchel–Young gradient is $p - e_y$; the exact likelihood gradient is $-J e_y / p_y$, which routes the label through its photo-finish edges. They coincide exactly when the edge weights are $p_i p_j$, i.e. for softmax (Section 6).
5. **A covariance-learning caution.** The heat identity $\partial G / \partial D_i = J_{ii} / 2 > 0$ and $\nabla_V G = JV$ make the Fenchel–Young loss monotone in choice-relevant perturbation variance: it cannot identify free Gaussian noise (Section 6).

Every displayed identity is verified numerically against finite differences, Monte Carlo, or direct linear algebra (Table 1); the neural experiments of Section 7 are pending.

2 The structured Gaussian perturbed maximum

Conditional on f , the scores are independent with means $m_i(f) = \mu_i + v_i^\top f$ and variances D_i . Write $F_i(x | f)$ and $g_i(x | f)$ for the conditional CDF and density. Then

$$p_i = \mathbb{P}(U_i = \max_j U_j) = \mathbb{E}_f \int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx.$$

The shared field is $H_f(x) = \prod_j F_j(x | f)$; each winner integrand divides one factor back out. On Q factor nodes and an L -point lattice, all N probabilities cost $O(QNL)$ (Cotton, 2026).

3 The potential from the same field

Conditional on f , H_f is the CDF of the maximum, so for the lattice endpoint b ,

$$\mathbb{E}[\max_i U_i | f] = b - \int_{-\infty}^b H_f(x) dx,$$

and $G(\mu)$ is the factor average. The forward pass has already built H_f at every node and lattice point; the potential is one more reduction.

One discretization point is not cosmetic. The continuum identity $\nabla G = p$ survives the lattice only if the potential’s quadrature is consistent with the probability quadrature: a rectangle-rule potential next to the implemented probability integrals broke the identity at 2.5×10^{-4} in our tests, while the trapezoidal form restores it to 3.6×10^{-8} . Fenchel–Young training needs a potential-consistent mode, with the discrete $\nabla G = p$ defect reported, not assumed away.

4 Dual geometry

G is convex and $G(\mu + c\mathbf{1}) = G(\mu) + c$, so the conjugate $\Psi = G^*$ lives on the simplex and, on the mean-zero gauge, $p = \nabla G(\mu)$ and $\mu = \nabla \Psi(p)$: the share inversion of the companion paper is the mirror map, and $\Psi(p) = p^\top \mu(p) - G(\mu(p))$ is computable from the inverse transform plus the same-pass potential.

The Hessian is the photo-finish Laplacian

$$J = \nabla^2 G = \sum_{i < j} w_{ij} (e_i - e_j)(e_i - e_j)^\top, \quad w_{ij} = \mathbb{E}_f \int g_i g_j \prod_{\ell \neq i, j} F_\ell dx > 0,$$

with w_{ij} the density of an i – j photo finish for the winning position. On the tangent space, $\nabla^2 \Psi(p) = J^+$. For the transfer direction $d_{ij} = e_i - e_j$,

$$d_{ij}^\top J^+ d_{ij} = \text{effective resistance between } i \text{ and } j$$

in the network with conductances w_{ab} : probability moves cheaply between strongly competing alternatives and expensively across weak competitive pathways.

Softmax is the special case $w_{ij} = p_i p_j$, whose resistance is $1/p_i + 1/p_j$ — exactly the curvature of negative Shannon entropy along d_{ij} . Shannon geometry is the effective-resistance geometry of one particular competition graph. A structured Gaussian layer separates information softmax cannot: two states with identical p can carry different photo-finish graphs, distinguishing “similar marginal probability” from “direct competitors at the decision boundary”.

5 Shared-field reverse mode

The companion paper differentiates p in μ . A heteroscedastic layer needs V and D as well. Let $r_i(x, f) = g_i \prod_{\ell \neq i} F_\ell$ be the winner densities, let a downstream loss supply the cotangent $c_i = \partial \mathcal{L} / \partial p_i$, and form one additional field $S_c = \sum_i c_i r_i$ with hazards $\lambda_j = g_j / F_j$.

Proposition 1 (Shared-field vector–Jacobian product). *With $a_j(f) = \int [c_j r_j \frac{x - m_j}{D_j} - (S_c - c_j r_j) \frac{x - m_j}{2D_j} \lambda_j] dx$ and $b_j(f) = \int [c_j r_j \frac{(x - m_j)^2 - D_j}{2D_j^2} - (S_c - c_j r_j) \frac{x - m_j}{2D_j} \lambda_j] dx$,*

$$\frac{\partial \mathcal{L}}{\partial \mu_j} = \mathbb{E}_f[a_j], \quad \frac{\partial \mathcal{L}}{\partial v_j} = \mathbb{E}_f[a_j f], \quad \frac{\partial \mathcal{L}}{\partial D_j} = \mathbb{E}_f[b_j].$$

All N parameter blocks reuse the same S_c ; the cost is $O(QN(L+k))$.

Proof. Only two terms depend on alternative j 's parameters: its own winner density through g_j , and every other winner density through F_j . Hence $\partial_{\theta_j} \mathcal{L} = \mathbb{E}_f \int [c_j r_j \partial_{\theta_j} \log g_j + (S_c - c_j r_j) \partial_{\theta_j} \log F_j] dx$. For a Gaussian factor, $\partial_{m_j} \log g_j = (x - m_j)/D_j$ and $\partial_{m_j} \log F_j = -\lambda_j$; $\partial_{D_j} \log g_j = ((x - m_j)^2 - D_j)/(2D_j^2)$ and $\partial_{D_j} \log F_j = -\lambda_j(x - m_j)/(2D_j)$. Finally $\partial_{\mu_j} m_j = 1$ and $\partial_{v_j} m_j = f$. $\square$

Measured against central finite differences (experiment 27), the three blocks agree to 2.2×10^{-9} , 4.9×10^{-10} , and 6.5×10^{-10} .

If the implementation normalizes, $p = \tilde{p}/s$ with $s = \mathbf{1}^\top \tilde{p}$, the incoming cotangent must first be pulled back: $\tilde{c} = (c - (c^\top p)\mathbf{1})/s$. Differentiating the unnormalized field while treating normalization as constant is incorrect.

6 Two learning objectives

Fenchel–Young. For a one-hot label e_y and centered Gaussian noise, $\Psi(e_y) = 0$, so the Fenchel–Young loss induced by G is

$$L_{\text{FY}}(\mu, y) = G(\mu) - \mu_y, \quad \nabla_\mu L_{\text{FY}} = p - e_y,$$

the exact Gaussian counterpart of the softmax cross-entropy gradient. In the binary case, with $\delta = \mu_1 - \mu_2$ and $s^2 = \text{Var}(\xi_1 - \xi_2)$, $L_{\text{FY}} = s[\phi(\delta/s) - (\delta/s)\Phi(-\delta/s)] = \mathbb{E}[(U_2 - U_1)_+]$: a Gaussian-smoothed hinge whose training signal concentrates on photo finishes (verified to 4.9×10^{-13} against the field computation).

Likelihood. Treating the race as a probabilistic model gives $L_{\text{NLL}} = -\log p_y$, and by symmetry of J ,

$$\nabla_\mu L_{\text{NLL}} = -\frac{J e_y}{p_y} = \frac{1}{p_y} \sum_{j \neq y} w_{yj} (e_j - e_y) :$$

the label updates only through its photo-finish edges. For softmax, $w_{yj} = p_y p_j$ collapses this to $p - e_y$: the two objectives coincide exactly in the Gumbel/softmax case and separate otherwise. Their curvatures likewise separate: Fenchel–Young curvature is J ; the Fisher information of the winner likelihood is $J \text{diag}(p)^{-1} J$, and softmax satisfies $J \text{diag}(p)^{-1} J = J$.

Covariance collapse. The Gaussian potential obeys the heat identity $dG = \frac{1}{2} \text{tr}(J d\Sigma)$, hence

$$\frac{\partial G}{\partial D_i} = \frac{J_{ii}}{2} > 0, \quad \nabla_V G = JV$$

(both verified to $\sim 10^{-8}$). Since $-\mu_y$ carries no covariance dependence, $\nabla_{V,D} L_{\text{FY}} = \nabla_{V,D} G$: the Fenchel–Young loss is monotone increasing in every positive-semidefinite direction of choice-relevant perturbation covariance. Unconstrained minimization removes the noise. If Σ is a chosen smoothing geometry, fix or constrain it; if $\Sigma(x)$ is uncertainty to be learned, use the winner likelihood or an outer criterion.

identity	max error
$\nabla_\mu G = p$	3.6×10^{-8}
reverse mode, μ block	2.2×10^{-9}
reverse mode, V block	4.9×10^{-10}
reverse mode, D block	6.5×10^{-10}
heat identity $\partial G / \partial D_i = J_{ii} / 2$	1.0×10^{-8}
$\nabla_V G = JV$	7.7×10^{-9}
$\nabla_\mu (-\log p_y) = -J e_y / p_y$	5.1×10^{-9}
binary Fenchel–Young closed form	4.9×10^{-13}
dual curvature = effective resistance	1.4×10^{-6} (rel.)
softmax $R_{ij} = 1/p_i + 1/p_j$	1.8×10^{-15}
G against 4×10^6 -draw Monte Carlo	1.9×10^{-4} ($\frac{1}{2}$ s.e.)

Table 1: Every claimed identity, verified (experiment 27: $N = 5$, $k = 1$, Gauss–Hermite 21, $L = 2001$, against central finite differences, Monte Carlo, or direct linear algebra).

Gauges. An end-to-end layer must remove three invariances: $\mu \mapsto \mu + c\mathbf{1}$; $V \mapsto V + \mathbf{1}a^\top$ (common factor noise); and joint scaling $(\mu, V, D) \mapsto (s\mu, sV, s^2D)$. Project μ and V onto mean-zero columns and normalize the contrast covariance, e.g. $\text{tr}(P\Sigma P)/(N - 1) = 1$ with $P = I - \mathbf{1}\mathbf{1}^\top/N$. Factor rotation $V \mapsto VR$ remains unidentified and is harmless for prediction.

7 Experimental program

Three questions, kept separate. **A** (numerical): can the operator be evaluated and differentiated accurately without resampling? **B** (optimization): does a fixed accurate gradient beat M -sample Monte Carlo gradients at matched wall time — including $M = 1$, which Berthet et al. (2020) note is often adequate for SGD? **C** (statistical): does structured Gaussian competition improve the model? A faster operator is not evidence of a better inductive bias, and vice versa.

The planned sequence: (I) operator microbenchmarks over $N \in \{10, \dots, 10^4\}$, $k \leq 15$ (product Gauss–Hermite at low k , fixed scrambled Sobol above), reporting probability, potential, and gradient error against wall time and memory, with equal-time and equal-error comparisons against hard-argmax Monte Carlo, common-random-number and RQMC variants, and the temperature-softmax surrogate; (II) the identity suite of Table 1 extended to automated randomized tests plus autograd **gradcheck**; (III) frozen-feature classification (convex in the head) comparing softmax, Gaussian Fenchel–Young at $M \in \{1, \dots, 1000\}$, the resampling-free evaluator, and exact winner NLL, on objective gap and accuracy versus wall clock; (IV) replication of the Gaussian CIFAR-10 experiment of Berthet et al. (2020) with one added treatment, the deterministic evaluator; (V) structured fixed covariance from held-out confusion, prototypes, or augmentation logit covariance at $k \in \{1, 2, 4, 8\}$; (VI) the heteroscedastic classifier of Collier et al. (2021) re-trained with the exact winner likelihood via Proposition 1, against the Monte-Carlo temperature-softmax surrogate, both evaluated under a common high-accuracy reference.

Success requires any of: materially better accuracy-per-wall-time than Monte Carlo; materially lower gradient error at matched time; faster convergence to a validation threshold; large latency or memory savings at high accuracy; or better-calibrated models from the exact likelihood. The project pivots if $M = 1$ Monte Carlo trains as well and faster, if useful ranks

are unreachable, or if structure adds no statistical gain — in which case the operator’s role is high-accuracy inference and calibration, not SGD replacement.

8 Discussion

Softmax is the explicitly computable Gumbel member of the perturbed-max family. The results above make the low-rank Gaussian member explicit too: potential, probabilities, inverse, and all parameter gradients from one shared field, with a graph geometry that softmax collapses to a single special case. The claim is not that Gaussian perturbations dominate Gumbel ones; it is that in this structured regime, computability no longer decides the comparison in advance. The heteroscedastic classification literature already states the demand: its generative model is this model, and its current evaluator is a sampled relaxation of it.

Reproducibility

Experiment 27 in the companion repository implements the operator and verifies every identity in Table 1; the companion paper’s experiments 13–26 cover the underlying transform, its certification, and its benchmarks.

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